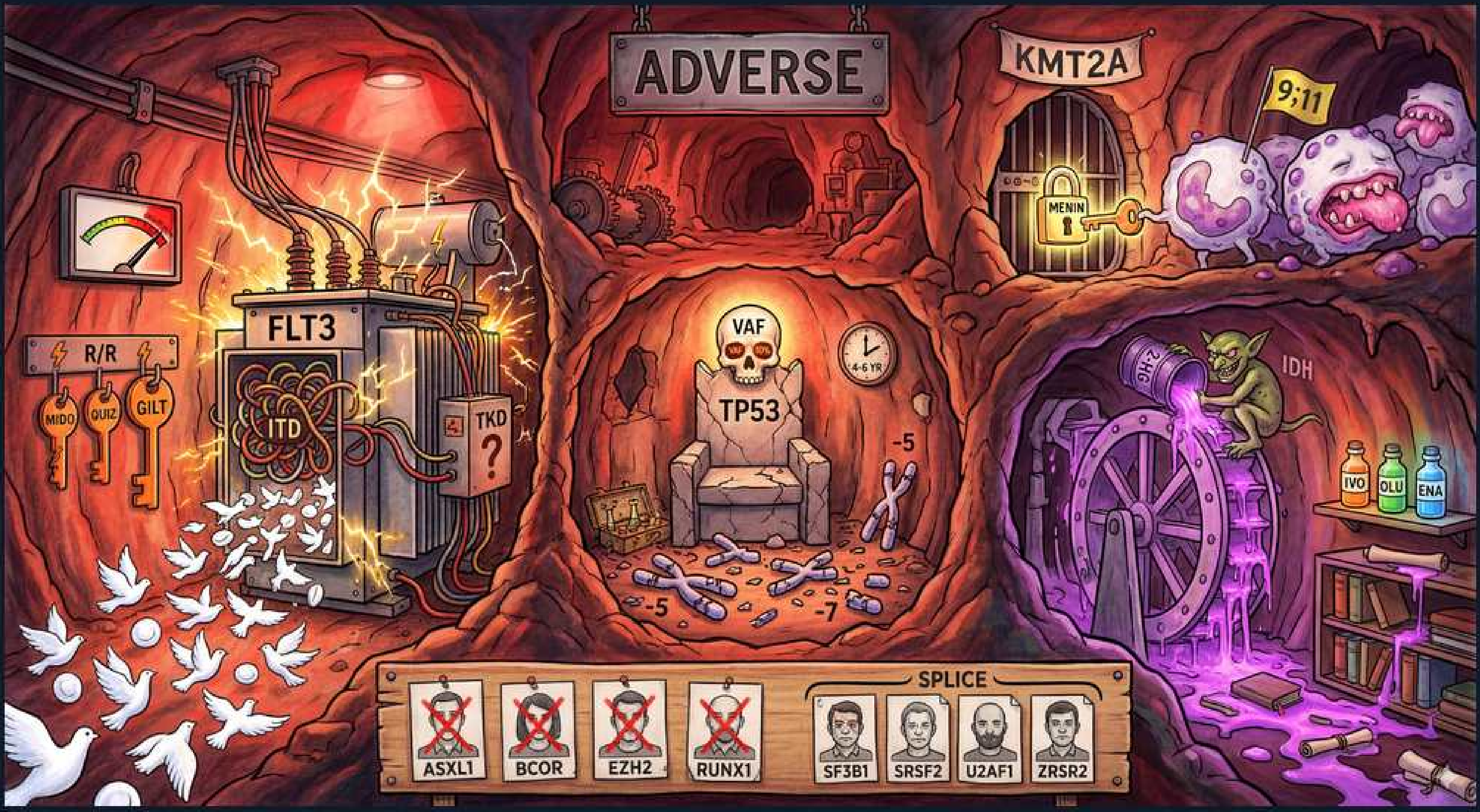
AML — Adverse-Risk Genetics (TP53, FLT3, KMT2A, Splice)



1. ADVERSE archway

WHAT YOU SEE

A red cave with an 'ADVERSE' archway over the whole scene.

WHAT IT ENCODES

This scene depicts ELN adverse-risk AML — the genetic features predicting poor response to standard chemotherapy, high relapse risk, and indication for allogeneic stem-cell transplant in first remission. Adverse risk includes TP53 mutation, complex/monosomal karyotype, KMT2A rearrangements, FLT3-ITD (high allelic ratio context), and certain secondary-type splicing mutations.

BOARD TRAP

Adverse-risk AML usually mandates allogeneic transplant in CR1 — favorable-risk [t(8;21), inv(16), NPM1 without FLT3-ITD] generally does NOT; misclassifying risk drives the wrong transplant decision.

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2. FLT3-ITD generator

WHAT YOU SEE

A crackling electrical transformer labeled 'FLT3' with an 'ITD' coil sparking, and a 'TKD?' switch.

WHAT IT ENCODES

FLT3 is a receptor tyrosine kinase; internal tandem duplication (FLT3-ITD) drives constitutive proliferative signaling, high WBC, and high relapse risk — adverse especially at high allelic ratio. FLT3 inhibitors (midostaurin added to 7+3 induction; gilteritinib for relapsed/refractory) improve outcomes. FLT3-TKD (D835) is a separate point mutation.

BOARD TRAP

Add midostaurin to induction for FLT3-MUTATED AML (ITD or TKD), and use gilteritinib in relapse — FLT3-ITD with high allelic ratio shifts NPM1-mutant AML from favorable toward adverse risk.

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3. R/R (relapsed-refractory) keys

WHAT YOU SEE

Keys labeled 'R/R', 'MIDO', 'GILT' near the FLT3 transformer.

WHAT IT ENCODES

Therapeutic 'keys' for FLT3 disease: midostaurin (first-generation, with induction/consolidation) and gilteritinib (potent second-generation, for relapsed/refractory FLT3-mutated AML, superior to salvage chemo in the ADMIRAL trial). Quizartinib is another FLT3-ITD-selective inhibitor added to chemo.

BOARD TRAP

Gilteritinib monotherapy beats salvage chemotherapy in relapsed/refractory FLT3-mutated AML — don't default to intensive salvage chemo when a targeted FLT3 inhibitor is indicated.

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4. TP53 throne (VAF)

WHAT YOU SEE

A central throne with a 'TP53' skull and 'VAF' crown, surrounded by '-5' and '-7' chromosome cards.

WHAT IT ENCODES

TP53 mutation (especially multi-hit or VAF ≥10%) is the SINGLE WORST prognostic lesion in AML/MDS — chemo-resistant, frequently with complex/monosomal karyotype (loss of chromosomes 5/7, i.e., -5/del(5q), -7/del(7q)). Outcomes are poor even with allogeneic transplant; clinical trials are preferred.

BOARD TRAP

TP53-mutated AML is the most adverse subtype and resists standard chemo AND venetoclax-based regimens over time — even transplant outcomes are poor; favor trials. Don't expect favorable-risk-level response.

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5. Complex karyotype (−5 / −7 cards)

WHAT YOU SEE

Scattered cards reading '-5' and '-7' (monosomies/deletions) around the TP53 throne.

WHAT IT ENCODES

Complex karyotype (≥3 unrelated chromosomal abnormalities) and monosomal karyotype, frequently involving -5/del(5q) and -7/del(7q), are adverse-risk and strongly associated with TP53 mutation and therapy-related/secondary AML. They predict chemoresistance and short survival.

BOARD TRAP

Monosomy 5 and 7 (and complex karyotype) cluster with TP53 and secondary/therapy-related AML — all adverse; these are NOT the favorable core-binding-factor abnormalities.

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6. KMT2A (MLL) locked gate / t(9;11)

WHAT YOU SEE

A locked gate labeled 'KMT2A' with a key, and a '9;11' tag on monstrous blasts.

WHAT IT ENCODES

KMT2A (MLL, 11q23) rearrangements — e.g., t(9;11)(MLLT3::KMT2A) and many partner translocations — are generally adverse/intermediate-adverse, common in therapy-related AML (after topoisomerase-II inhibitors like etoposide) and in monocytic/infant leukemias. Menin inhibitors (e.g., revumenib) are emerging targeted therapy.

BOARD TRAP

KMT2A/11q23-rearranged AML is classically therapy-related from TOPOISOMERASE-II inhibitors (etoposide/anthracyclines) with a SHORT latency — distinct from alkylator-induced -5/-7 t-AML with a longer latency.

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7. Monocytic / IDH-driven blasts

WHAT YOU SEE

Purple monstrous blasts with 'IDH' near the KMT2A gate.

WHAT IT ENCODES

IDH1/IDH2 mutations produce the oncometabolite 2-hydroxyglutarate, blocking myeloid differentiation. Targeted inhibitors: ivosidenib (IDH1) and enasidenib (IDH2) induce differentiation and can cause differentiation syndrome. IDH status is intermediate-risk but therapeutically actionable.

BOARD TRAP

IDH1/IDH2 mutations are actionable with ivosidenib/enasidenib (differentiation agents) — and these too can cause a differentiation syndrome (like ATRA in APL); recognize the targeted option.

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8. IDH inhibitor potions

WHAT YOU SEE

Colored potion bottles on a shelf beside the IDH blasts.

WHAT IT ENCODES

IDH-directed 'potions' (ivosidenib for IDH1, enasidenib for IDH2) are oral differentiation therapies used in relapsed/refractory and unfit newly diagnosed AML (ivosidenib + azacitidine). They restore myeloid maturation rather than directly killing blasts.

BOARD TRAP

IDH inhibitors work by DIFFERENTIATION, not cytotoxicity — like APL therapy they can trigger differentiation syndrome and a transient rise in counts; don't mistake that for progression.

9. ASXL1 / BCOR / EZH2 / RUNX1

WHAT YOU SEE

A row of mugshot cards: ASXL1, BCOR, EZH2, RUNX1 under a 'wanted' frame.

WHAT IT ENCODES

ASXL1, BCOR, EZH2, RUNX1 (and STAG2, U2AF1, SRSF2, SF3B1, ZRSR2) are 'secondary-type' mutations marking AML arising from prior MDS/MPN (secondary AML) — an adverse-risk group with chromatin/transcription-factor and spliceosome involvement. RUNX1-mutated AML is specifically adverse.

BOARD TRAP

Secondary-type mutations (ASXL1, BCOR, EZH2, RUNX1, STAG2) define AML with MDS-related/secondary biology = adverse risk — even when the karyotype looks unremarkable; they push toward transplant.

10. Spliceosome mutations (SF3B1/SRSF2/U2AF1/ZRSR2)

WHAT YOU SEE

A 'SPLICE' panel with mugshots SF3B1, SRSF2, U2AF1, ZRSR2.

WHAT IT ENCODES

Spliceosome gene mutations (SRSF2, SF3B1, U2AF1, ZRSR2) disrupt RNA splicing and are hallmarks of MDS-related/secondary AML. SRSF2 and U2AF1 are adverse secondary-type lesions; SF3B1 is classically tied to ring sideroblasts in MDS. Their presence reclassifies AML as MDS-related (adverse).

BOARD TRAP

Spliceosome mutations signal MDS-related/secondary AML (adverse) — SF3B1 specifically links to ring sideroblasts in MDS; don't read these as favorable just because the blast count or karyotype looks tame.

11. Doves (low blast / cytopenia)

WHAT YOU SEE

White doves/birds scattering from the FLT3 transformer base.

WHAT IT ENCODES

The fleeing white cells evoke the cytopenias and bone-marrow failure of high-risk/secondary AML — pancytopenia from marrow replacement and dysplastic, ineffective hematopoiesis, a frequent presenting picture in MDS-related AML.

BOARD TRAP

Cytopenias with dysplasia preceding overt AML point to MDS-related/secondary disease (adverse) rather than de novo favorable-risk AML — the antecedent history reframes risk.

12. VAF clock (clonal burden)

WHAT YOU SEE

A clock and 'VAF' sign on the TP53 throne.

WHAT IT ENCODES

Variant allele frequency (VAF) quantifies mutational clonal burden; for TP53, a higher VAF (and multi-hit/biallelic status) defines the most adverse category. VAF also informs measurable residual disease tracking and clonal evolution at relapse.

BOARD TRAP

TP53 risk depends on allelic state and VAF — multi-hit/biallelic or high-VAF TP53 is the worst; a single low-VAF hit may behave less aggressively. The molecular detail, not just 'TP53 present', drives prognosis.

13. Allo-transplant indication (CR1)

WHAT YOU SEE

The fortified 'ADVERSE' cave implying escalation of therapy.

WHAT IT ENCODES

Adverse-risk genetics are the principal indication for allogeneic hematopoietic stem-cell transplant in first complete remission, since chemotherapy alone yields high relapse rates. Donor search should begin early. For unfit patients, azacitidine + venetoclax is a common lower-intensity backbone.

BOARD TRAP

The reflex for adverse-risk AML is allo-transplant in CR1 (begin donor search at diagnosis) — and for unfit patients, aza+venetoclax; standard 7+3 alone is insufficient for cure in this group.

14. Venetoclax resistance (TP53)

WHAT YOU SEE

Pills scattered at the foot of the TP53 throne (failed therapy).

WHAT IT ENCODES

While azacitidine + venetoclax is highly active in many older AML genotypes (IDH, NPM1), TP53-mutated AML responds poorly and relapses quickly even on venetoclax-based therapy. This resistance underscores why TP53 is the worst category and trials/novel agents are preferred.

BOARD TRAP

Venetoclax + azacitidine is NOT a durable answer for TP53-mutated AML (early resistance) — its best responses are in IDH-mutated and NPM1-mutated disease; don't assume venetoclax rescues TP53.